

Archetype 02: Comparison Tables Corpus

This document contains 10 comparison tables evaluating products, tiers, algorithms, protocols, and vehicle benchmarks across multiple feature columns.

Table 1: Cloud Storage Service Comparison

Feature	AWS S3	Google Cloud Storage	Azure Blob Storage
Standard Storage (\$/GB)	\$0.023	\$0.020	\$0.018
SLA Uptime	99.99%	99.95%	99.90%
Versioning Support	Yes	Yes	Yes
Object Locking	Yes (WORM)	Yes (Bucket Lock)	Yes (Immutability)

Table 2: LLM Framework Evaluation

Dimension	LangChain	LlamaIndex	Haystack	DSPy
Primary Focus	Agent Workflows	Data Indexing / RAG	Production NLP	Programmatic Prompts
Learning Curve	Moderate	Low-Moderate	Moderate	Steep
Graph Support	GraphQA / Cypher	Knowledge Graphs	Pipeline Graphs	Modular Chains
Active Community	Very High	High	High	Growing

Table 3: Vector Database Performance Benchmarks

Metric	Pinecone	Milvus	Qdrant	ChromaDB
QPS (1M vectors, 768d)	1,200	2,450	2,100	450
P99 Latency (ms)	12.4	8.1	9.2	34.0
HNSW Indexing Time	Managed	14 mins	11 mins	35 mins
Hybrid Search (Sparse+Dense)	Yes	Yes	Yes	Partial

Table 4: Subscription Plan Tiers

Plan Feature	Free Starter	Professional	Enterprise Scale
Monthly Price	\$0	\$49 / mo	\$499 / mo
API Requests / day	1,000	100,000	Unlimited
Dedicated Support	Community	Email (24h)	24/7 Phone & Slack SLA
SSO / SAML	No	No	Yes

Table 5: Programming Language Concurrency Models

Language	Concurrency Primitive	Runtime Overhead	Memory Safety Guarantee
Rust	Async/Await + OS Threads	Zero-cost Abstraction	Compile-time guaranteed
Go	Goroutines + Channels	Low (2KB stack)	Garbage collected
Python	Asyncio / Threading / MP	GIL Bottleneck	Ref-counted GC
Java	Virtual Threads (Project Loom)	Low Carrier Threading	JVM Managed

Table 6: Relational Database Feature Matrix

Capability	PostgreSQL	MySQL 8.0	Oracle 21c	SQL Server 2022
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JSON Query Support	JSONB (GIN Index)	JSON Functions	JSON Datatype	JSON Path Functions
Full Text Search	Built-in tsvector	InnoDB FTS	Oracle Text	Full-Text Engine
Window Functions	Full SQL:2016	Standard Support	Advanced Analytic	Full Standard Support

Table 7: Mobile Operating System Security Comparison

Security Control	Apple iOS 18	Google Android 15	GrapheneOS
App Sandboxing	Strict Seatbelt profile	SELinux & UID sandbox	Hardened Sandbox + Exec Spawning
Biometric Enclave	Secure Enclave (SEP)	Titan M2 / StrongBox	Hardware TrustZone / Titan M2
Zero-Day Exploit Mitigation	BlastDoor / Lockdown Mode	Memory Tagging (MTE)	Hardened Malloc + Exploit Protections

Table 8: Electric Vehicle Model Specs Comparison

Specification	Tesla Model S Plaid	Porsche Taycan Turbo GT	Lucid Air Sapphire
0-60 mph (sec)	1.99 s	2.1 s	1.89 s
Top Speed (mph)	200 mph	190 mph	205 mph
EPA Range (miles)	359 mi	280 mi	427 mi
Peak DC Charging (kW)	250 kW	320 kW	300 kW

Table 9: Message Broker Architecture Matrix

Broker	Architecture Type	Max Throughput	Ordering Guarantee	Storage Retention
Apache Kafka	Distributed Commit Log	> 1,000,000 msg/s	Strict Per-Partition	Time/Size Based Disk
RabbitMQ	AMQP Smart Broker	50,000 msg/s	Strict Per-Queue FIFO	Transient / Queue ack
Apache Pulsar	Tiered BookKeeper Log	> 800,000 msg/s	Per-Partition Key	Tiered S3 / Cloud Cold
NATS JetStream	Raft-replicated Stream	> 5,000,000 msg/s	Per-Subject Stream	Configurable limits

Table 10: Sorting Algorithm Complexity Comparison

Algorithm	Best Time	Average Time	Worst Time	Space Complexity
Quicksort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$	$O(\log n)$
Mergesort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$
Heapsort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(1)$
Timsort	$O(n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$